Electric Fields of Static and Moving Charges
Course Code: USPHY5002CR1

Analytical Study of Damping and Parasitic Resistance in a Series LRC Circuit

Name: Samuel Gardonis
Roll no: 219
UID: 2402070

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Abstract

The objective of this experiment was to investigate the transient behavior of a series L–R–C electrical circuit and to characterize the damping of oscillations arising from resistive energy dissipation. When a square-wave voltage is applied to such a circuit, the capacitor repeatedly charges and discharges through the inductor and resistor, producing a transient oscillatory response. In practical circuits, resistance causes gradual loss of stored electromagnetic energy, resulting in damped harmonic oscillations whose amplitude decreases exponentially with time.

The experiment involved constructing a series L–R–C circuit using a fixed capacitor and inductor while systematically varying the external resistance. A square-wave signal from a function generator was used to excite the circuit, and the resulting voltage waveform across the capacitor was recorded using a digital storage oscilloscope. Successive peak amplitudes and oscillation periods were measured to determine the logarithmic decrement and the damping constant for each resistance value.

The dependence of the damping constant on total circuit resistance was analyzed through linear regression. From the slope and intercept of the resulting plot, the inductance of the coil and its internal resistance were estimated. The results demonstrated a linear relationship between damping and resistance, consistent with theoretical predictions derived from the differential equation governing L–R–C circuits. The presence of a non-zero intercept in the regression confirmed the existence of internal resistance within the inductor windings. Overall, the experimental observations were found to be in reasonable agreement with theoretical expectations for underdamped electrical oscillations.

Introduction

Oscillatory behavior is a fundamental phenomenon observed in many physical systems, ranging from mechanical vibrations to electromagnetic circuits. Electrical oscillations in L–R–C circuits provide a direct analogue to mechanical harmonic oscillators such as mass–spring systems. In such circuits, energy alternates between the electric field stored in a capacitor and the magnetic field associated with current flowing through an inductor. In an ideal system with no resistance, this exchange of energy would continue indefinitely, producing sustained oscillations at a constant amplitude.

In real circuits, however, resistive elements are always present. Resistance causes electrical energy to be dissipated as heat through Joule heating, which gradually reduces the amplitude of oscillations. As a result, the circuit exhibits **damped harmonic motion**, where the oscillation amplitude decreases exponentially with time. The rate of decay is determined by the damping constant, which depends on the total resistance present in the circuit.

The theoretical behavior of a series L–R–C circuit can be obtained using Kirchhoff's loop rule. Considering a circuit containing an inductor (L), resistor (R), and capacitor (C), the governing differential equation for charge (q) on the capacitor is

$$\frac{d^2q}{dt^2} / + \frac{R}{L}\frac{dq}{dt} + \frac{q}{LC} = 0$$

This second-order linear differential equation is mathematically identical to that describing a damped mechanical oscillator. Its solution predicts oscillatory motion when the resistance is sufficiently small, specifically when

$$\frac{R}{2L} < \frac{1}{\sqrt{LC}})$$

Under this **underdamped condition**, the voltage across the capacitor varies according to

$$V(t) = V_0 e^{-\alpha t} cos(\omega t + \phi)$$

where
α = R/2L is the damping constant and
ω is the damped angular frequency.

An important quantity used to quantify damping in oscillatory systems is the **logarithmic decrement**, which measures the rate at which successive amplitudes decrease. It is defined as

$$\delta = ln(\frac{A_1}{A_2})$$

Where A_1 and A_2 are consecutive peak amplitudes. From this value, the damping constant can be calculated by dividing the logarithmic decrement by the oscillation period.

In practical circuits, the total resistance is not limited to the external resistor inserted into the circuit. Additional resistance arises from the copper windings of the inductor, connecting wires, and internal components of measuring instruments. This inherent resistance is often referred to as **parasitic or internal resistance**. Determining its magnitude is important in many applications, particularly in resonant circuits and signal-processing systems where damping affects performance and frequency selectivity.

The present experiment aims to analyze the damping behavior of an L–R–C circuit by measuring the decay of oscillations produced by a square-wave excitation. By varying the external resistance and calculating the corresponding damping constants, the relationship between resistance and damping can be examined experimentally. A linear fit of damping constant versus resistance allows estimation of both the inductance of the coil and the internal parasitic resistance associated with the inductor. This analysis provides experimental verification of the theoretical model of damped electrical oscillations and highlights the influence of non-ideal components in real electrical circuits.

Damping Constant

$$\alpha = \frac{\delta}{T}$$

Damping Constant (Theoretical):

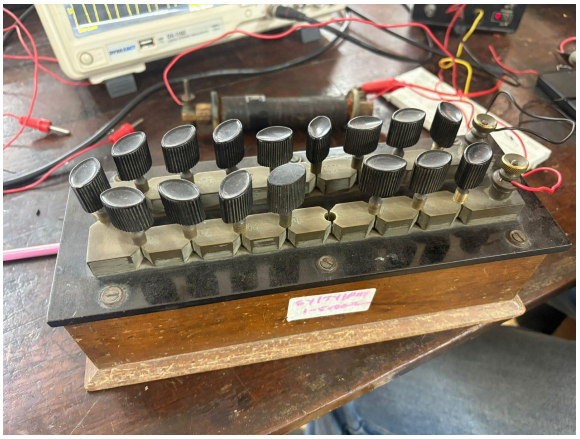
$$\alpha = \frac{R_{ext}+R_L}{2L}$$

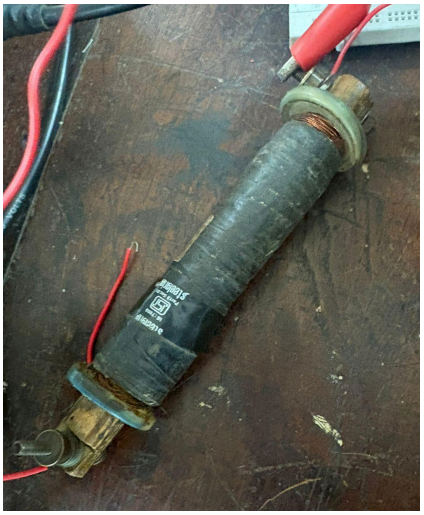
Procedure

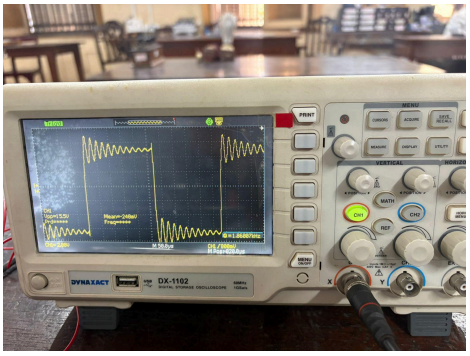
Assemble the series LCR circuit by connecting the inductor (L), capacitor (C), and variable resistor (R) in series. Connect a signal generator to provide a square-wave pulse that triggers the transient response. In parallel with the capacitor, connect a Digital Storage Oscilloscope (DSO) to monitor the voltage decay.

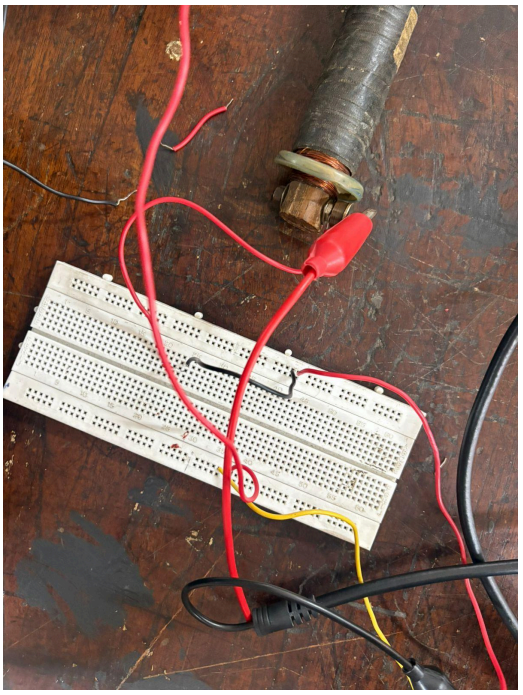
Adjust the resistance to observe underdamped oscillations on the DSO screen. For a specific R value, measure the amplitudes of successive peaks (A_n, A_{n+1}). Repeat this for various resistances. Finally, use the recorded amplitudes to calculate the logarithmic decrement delta and the damping coefficient beta, analyzing how varying resistance affects the decay rate.

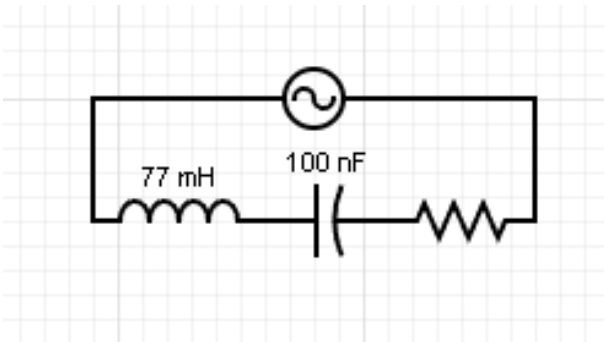
Setup and Circuit Diagram





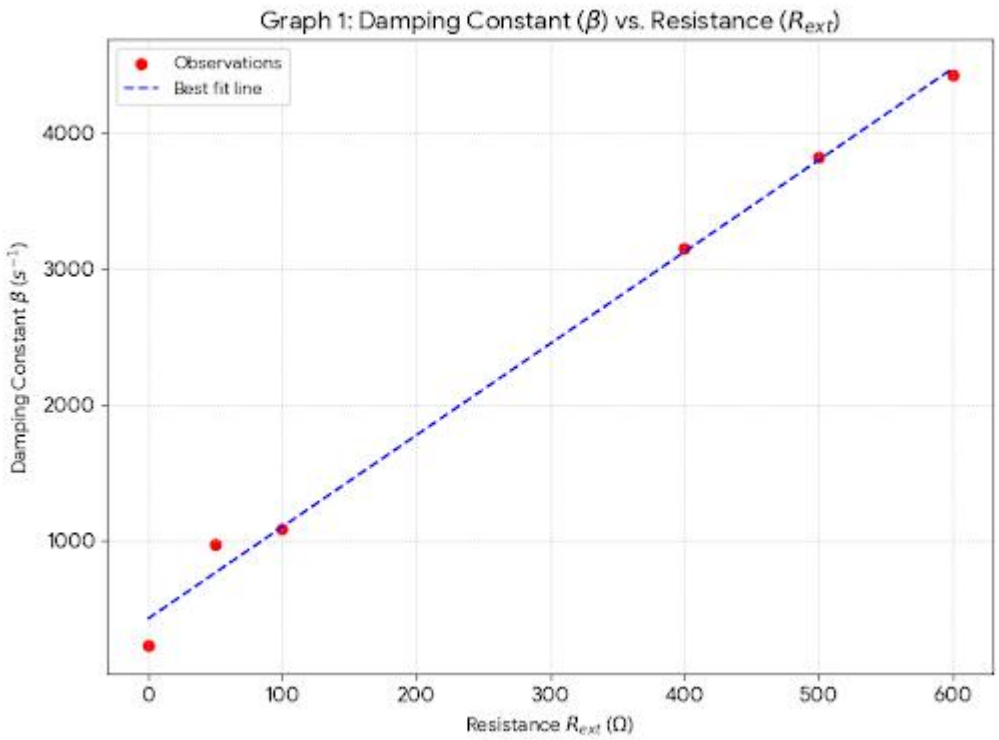


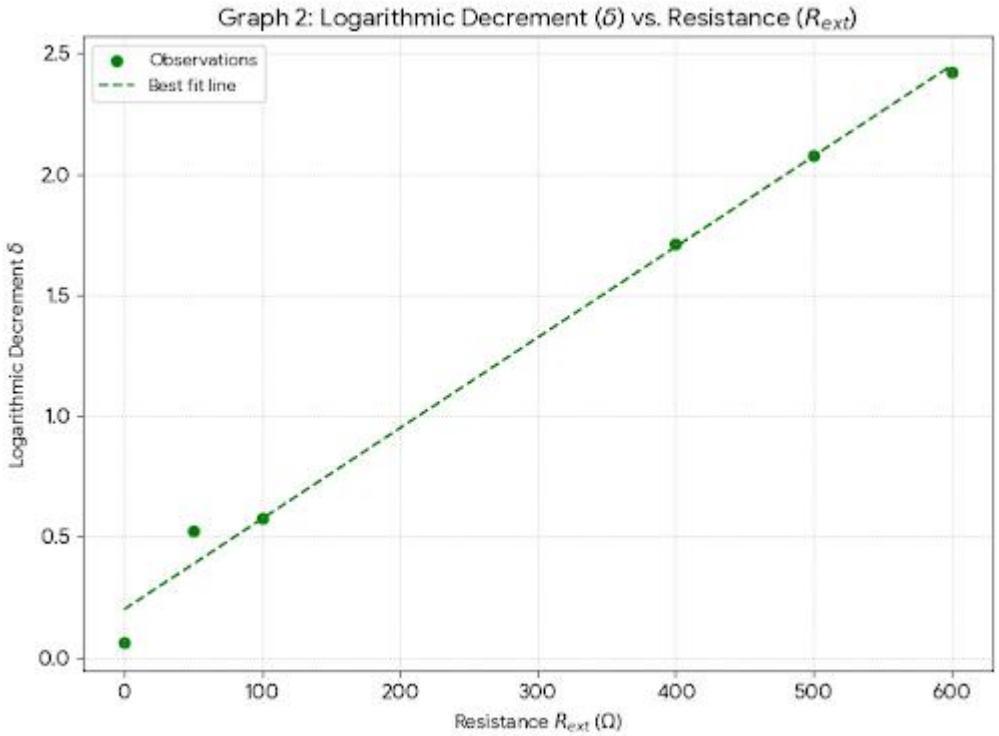


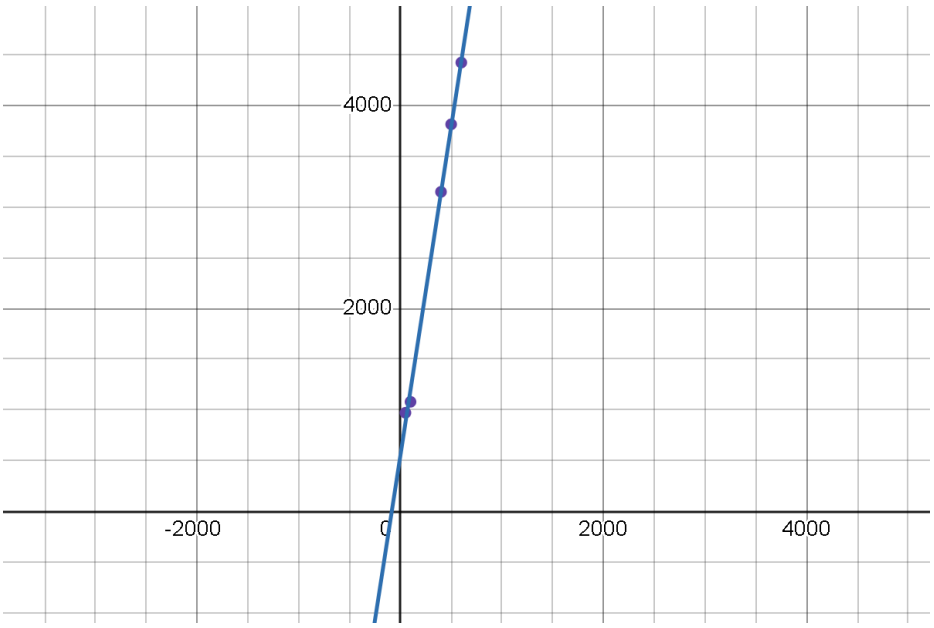


Observation Table & Graphs

R_{ext} (Ω)	Peak 1 A_1 (V)	Peak 2 A_2 (V)	Period T (μs)	Log Dec (δ)	Damping constant (δ) (s^{-1})	Q = π/δ
0	7.84	7.36	280	0.063	225.0	∞
50	12.70	7.52	540	0.524	970.4	6.0
100	9.84	5.52	536	0.578	1078.5	5.43
400	12.20	2.20	544	1.713	3148.9	1.83
500	10.40	1.30	545	2.079	3814.7	1.51
600	9.60	0.85	548	2.424	4423.4	1.30


Graph 1: Damping Constant (β) vs. Resistance (R_{ext})
Observations
Best fit line
Damping Constant β (s^{-1})
4000
3000
2000
1000
0
100
200
300
400
500
600
Resistance R_{ext} (Ω)


Graph 2: Logarithmic Decrement (δ) vs. Resistance (R_{ext})
Observations
Best fit line
Logarithmic Decrement δ
2.5
2.0
1.5
1.0
0.5
0.0
0
100
200
300
400
500
600
Resistance R_{ext} (Ω)



Graph Analysis:

Plotting α (y-axis) against R_{ext} (x-axis) yields a linear relationship: $\alpha = (1/2L)R_{ext} + R_L/2L$

- **Slope (m):** 6.46×10^{-3}
- **Intercept (c):** 548.2

Results/Inference

- **Inductance (L):** Derived from the slope

  (L = 1/2m)

  i.e L ≅ 77.4mH

- **Parasitic Resistance (R_L):** Derived from the intercept (R_L = c/m), R_L ≅ 84.9 **Ω**
- **Inference:** The non-zero y-intercept confirms the presence of parasitic resistance within the inductor windings. The system demonstrates that as R increases, the **Quality Factor (Q)**—the ratio of energy stored to energy dissipated—decreases, leading to a faster transition to a critically damped state.
- For small resistance (50–100 Ω) → Q≈5–6
- Oscillations decay slowly.
- The circuit behaves closer to an ideal resonant system.
- For large resistance (400–600 Ω) → Q≈1–2
- Strong damping.
- Oscillations die out quickly.
- This trend confirms the theoretical relation:

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{\omega_0}{2\alpha} = \frac{\pi}{\delta}$$

which shows quality factor decreases as resistance increases.

Applications

- **Filter Design:** Understanding damping is critical to the design of radio-frequency (RF) filters to ensure signal selectivity.
- **Automotive Ignition:** L-R-C circuits are used to generate the high-voltage sparks in engines; controlling the decay prevents electrode wear.
- **Sensor Calibration:** Identifying parasitic resistance is vital for the precision calibration of inductive proximity sensors.

Error Analysis

- **Component Tolerance:** The capacitor ("104") has a typical ± 10% tolerance, affecting the theoretical period calculation.
- **Frequency Dependence:** The skin effect in the inductor windings may cause R_L to vary slightly at different oscillation frequencies.
- **Instrumental Error:** The DSO sampling rate (1 GSa/s) provides high precision, but manual cursor placement on the peaks introduces a small human error margin (≅1-2\%).

References

1. Sears and Zemansky, University Physics with Modern Physics, 15th Edition – Chapter 31 (Alternating Current) and Section 14.7 (Damped Oscillations)
2. H.D. Young and R.A. Freedman – Standard LCR circuit analysis and damping theory